



AQUACULTURAL ENGINEERING

EDUCATION

| Year | Degree/Exam                         | Institute                                          | CGPA/Marks |
|------|-------------------------------------|----------------------------------------------------|------------|
| 2027 | M.TECH                              | IIT Kharagpur                                      | 8.91 / 10  |
| 2025 | Agricultural Engineering            | Dr. Panjabrao Deshmukh Krishi Vidyapeeth Akola     | 8.26 / 10  |
| 2019 | Higher Secondary School Certificate | Board of Secondary Education Madhya Pradesh Bhopal | 83%        |
| 2017 | High School Certificate             | Board of Secondary Education Madhya Pradesh Bhopal | 84.6%      |

PROJECTS

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| <b>SuryaVedha AI/ML Urban Heat Mitigation   BAH 2026   Finalist</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | <b>Jul 2026 - Aug 2026</b> |
| <ul style="list-style-type: none"><li>• Fused 250GB Landsat, Sentinel-2, ERA5 and CPCB data across 629K Delhi NCR pixels to build high-resolution urban heat prediction models.</li><li>• Compared RF, XGBoost and DNN; selected XGBoost with <math>R^2=0.856</math>, <math>RMSE=0.928^\circ C</math> hold-out test set, achieving best test performance.</li><li>• Applied SHAP to identify NDVI, air temperature and NDBI as key heat drivers, improving model interpretability.</li><li>• Generated 80 Pareto-optimal strategies using NSGA-II, balancing cost, cooling and equity, with solutions spanning <math>-1.1^\circ C</math> to <math>-2.9^\circ C</math>.</li></ul> |                            |
| <b>Drainage AI AI/ML-Based Drainage Network Design   MoPR IIT Tirupati Hackathon   Finalist</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | <b>Mar 2026</b>            |
| <ul style="list-style-type: none"><li>• Processed 1.5B+ LiDAR points from 10 villages across 5 states using PMF and Random Forest refinement to generate 2m-resolution DTMs.</li><li>• Built XGBoost + Isolation Forest ensemble for waterlogging hotspots, achieving mean F1 0.849 and cross-village AUC-ROC of 0.860 overall.</li><li>• Developed Attention U-Net spatial refinement with SHAP explainability, combining terrain features to improve hotspot prediction boundaries.</li><li>• Designed 747.9km drainage networks using outlet-connected MST, Manning's equation and Rational Method; simulated 69.7% flood reduction.</li></ul>                                |                            |
| <b>UYOM JAL AI-Powered Digital Twin for Climate Forecasting</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | <b>Jul 2026</b>            |
| <ul style="list-style-type: none"><li>• Built an AI-powered digital twin for India using satellite, climate and terrain data to forecast climate-driven water risks across regions.</li><li>• Developed a ConvLSTM-based spatiotemporal forecasting pipeline integrating multi-source environmental data for regional climate prediction.</li><li>• Added explainability and uncertainty quantification to forecasts, for confidence-aware interpretation and operational decision support</li><li>• Created an interactive dashboard for real-time forecasting, validation and scenario analysis to enable actionable decisions in practice.</li></ul>                          |                            |
| <b>CAPTCHA Character Recognition   Deep Learning</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | <b>Sep 2026</b>            |
| <ul style="list-style-type: none"><li>• Generated 50K synthetic CAPTCHA images with randomized fonts, colors, noise and geometric variations</li><li>• Trained a PyTorch CNN using 3232 grayscale inputs, affine/perspective augmentation, AdamW optimization and LR scheduling.</li><li>• Built an automated inference pipeline with character segmentation, softmax confidence scoring and GPU/CPU execution.</li><li>• Evaluated model performance using classification reports, confusion matrices, per-class accuracy and visual prediction analysis.</li></ul>                                                                                                             |                            |
| <b>ResearchMate   Local RAG Research Assistant</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | <b>Sep 2026</b>            |
| <ul style="list-style-type: none"><li>• Built a local RAG assistant for querying research PDFs using MiniLM embeddings and ChromaDB semantic retrieval</li><li>• Implemented PDF ingestion, recursive chunking and top-k retrieval with distance-based relevance filtering for grounded search</li><li>• Integrated Llama 3.2 via Ollama with Streamlit for context-grounded PDF Q&amp;A, indexing and page-level source display</li></ul>                                                                                                                                                                                                                                       |                            |

INTERNSHIPS

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| <b>Sundarban Chemicals Pvt. Ltd.   Kolkata   Business Development &amp; Design Engineer Intern</b>                                                                                                                                                                                                                                                                                                                                     | <b>May 2026 - July 2026</b> |
| <ul style="list-style-type: none"><li>• Mapped Pan-India water/wastewater sectors and developed a 450+ company B2B database covering treatment technologies</li><li>• Modelled electrochemical reactor systems in SolidWorks, creating 3D assemblies and component layouts for treatment systems</li><li>• Designed 4040 RO membrane housings and prepared 2D drawings of reactor components for manufacturing documentation</li></ul> |                             |

COMPETITION/CONFERENCE

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| <b>1. National Finalist - BAH 2026   NRSC, ISRO HYD   Top 34/15,104 Teams</b> | <b>2. MoPR IIT Tirupati Hackathon   Geospatial Intelligence   Top 38 Teams</b> |
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SKILLS AND EXPERTISE

**Programming:** Python, SQL, C/C++, Git/GitHub | **ML & Analytics:** Machine Learning, XGBoost, Random Forest, Isolation Forest, SHAP, Feature Engineering | **Deep Learning:** PyTorch, Attention U-Net, CNN, ConvLSTM | **Generative AI:** Generative AI, LLMs, LangChain, LangGraph, CrewAI, MCP, ChromaDB, LLM Evaluation | **Deployment:** FastAPI, Docker, Streamlit | **Data & Visualization:** NumPy, Pandas, Matplotlib, QGIS, GEE

POSITIONS OF RESPONSIBILITY

**Placement Co-ordinator** Aquacultural Engineering | **Manager**, Innovation and Incubation Cell, IIT KGP | **Secretary Library**, MMM Hall, IIT KGP | **Media Content Team Head**, GYWS, IIT KGP | **Prefect**, Raigarh Boys Hostel, Dr. PDKV Akola | **Executive Body Member**, Dr. PDCF Akola

EXTRA CURRICULAR ACTIVITIES

- **Annual Social Gathering 2025 | Steering Committee Member** | Coordinated event planning and execution, managing a 5+ lakh budget
- **National Service Scheme (NSS) | Volunteer** | Completed a 120-hour Special Camp, contributing to outreach and community activities
- **Served as Vice-Captain in SAEINDIA TIFAN 2025**, contributing to the design and development of an Automated Vegetable Transplanter

AWARDS AND ACHIEVEMENTS

- **Secured the highest CGPA** in B.Tech (Agricultural Engineering) 2025 graduating batch at Dr. PDKV, Akola
- **Felicited** by Dr. Panjabrao Deshmukh Competative Forum, Akola for academic achievement